

**MATH 2241 INTRODUCTION TO MATHEMATICAL ANALYSIS, FALL
SEMESTER 2026**

0.1. Basic information.

- Lecturer: Prof. Kei Yuen, CHAN
- Office: Run Run Shaw Building, Room 418
- Email: kychan1 [AT] hku.hk
- Lectures: Tuesday (MB201) 14:00-15:50 and Thursday 15:00-15:50 (MBG07)
- Consultation hours:

- Teaching Assistant: Mr. Jiatong, HE
- Office: Graduate House P206
- Email: u3011746 [AT] connect.hku.hk
- Consultation hours:

- Teaching Assistant: Ms. Yuhua, MENG
- Office: Run Run Shaw Building, Room 211A
- Email: yvette00 [AT] connect.hku.hk
- Consultation hours:

- Teaching Assistant: Mr. Cheng, SHOU
- Office: Graduate House P206
- Email: cshou [AT] connect.hku.hk
- Consultation hours:

- Course webpage: HKU Moodle
- Tutorial time: Tuesday 17:00-17:50 MW103; Wednesday 12:00-12:50 MB237; Wednesday 13:00-13:50 MB237

0.2. Topics covered.

- The real number system: the real numbers as an ordered field, supremum and infimum, the completeness axiom, denseness of the rational numbers.
- Sequences and series of real numbers: limits of sequences, properties of convergent sequences, monotone sequences and Cauchy sequences, subsequences, series, tests of convergence for series.
- Continuity of real-valued functions: properties of continuous functions, the extreme value theorem, the intermediate value theorem, uniform continuity, limits of functions.

- Differentiation: properties of differentiable functions, the mean value theorem, Taylor's theorem and its applications.
- Integration: construction of the Riemann integral using Darboux sums and Riemann sums, the fundamental theorem of calculus.

0.3. References.

- Lecture notes
- Reference books:
 - Robert G. Bartle, Donald R. Sherbert: *Introduction to Real Analysis* (Wiley, 2011, Fourth Edition)
 - Kenneth A. Ross: *Elementary Analysis: The Theory of Calculus* (Springer, 2013, Second Edition)

Some notes in progress may be uploaded for the convenience of students in following the lectures, but *students should download the updated and completed lecture notes after each chapter is finished.*

0.4. Prerequisite.

- Pass in MATH1013 or MATH1851 or MATH1821
- Basic knowledge of sets, functions and logic (e.g. ability to comprehend “for all”, “for some”, “if and only if” statements, etc.)

0.5. Assessments.

- (1) Assignments (10%)
- (2) Tests (40%)
- (3) Examination (50%)

0.6. **Calculators.** A calculator is not needed in the course. Calculators are **not** allowed in any tests or the examination.

0.7. **Expectations of students.** This is a difficult course designed for Mathematics major students and will focus on both conceptual and practical aspects. I expect students to work hard for the class, and to spend hours going through the details of the lecture notes and working on practice problems after lectures. Assignments and tests are usually more difficult than the examples covered in lectures. Successful students should be able to link the topics in the course and reorganize the subject at the end of the course.

0.8. **Assignments.** Assignments will be given every one to two weeks. Late submission of an assignment has to be approved by the teaching assistants. No late submissions are allowed after solutions are posted or discussed. Assignments will receive a score of zero if no submission is made.

The main purpose of assignments is to help students practice. Grading on assignments is usually milder than tests and the exam.

0.9. **Tentative schedule.** Teaching period: The following schedule is approximate and will be adjusted during the semester:

Week	Topic	Remarks
1/9-4/9	Real numbers	
7/9-11/9	Real numbers	
14/9-18/9	Sequences of numbers	
21/9-25/9	Sequences of numbers	
28/9-2/10	Sequences of numbers	1/10 (No class)
5/10-9/10	Continuity	8/10 (Test 1)
12/10-16/10		Reading Week
19/10-23/10	Continuity	
26/10-30/10	Continuity	
2/11-6/11	Differentiation	
9/11-13/11	Differentiation	
16/11-20/11	Integration	19/11 (Test 2)
23/11-27/11	Integration	

0.10. **Tests.** The two tests have equal weighting. Make-up tests will only be given in rare circumstances such as mandatory university events. Reasons such as personal or association commitments are not entertained. A written request with supporting documents for missing a test should be provided as soon as possible before a test.

0.11. **Plagiarism.** Students have to be aware of the consequences of plagiarism:
<https://tl.hku.hk/plagiarism/>

0.12. **Tutorials and lectures.** There will be a weekly tutorial starting from the week of 14 September. Active participation in tutorials and lectures is highly encouraged. In order to follow lectures or tutorials, you may read the lecture notes before coming to class and keep a question in mind. During the lecture, try to find the answer. If you cannot find the answer, ask the question in class or after class.